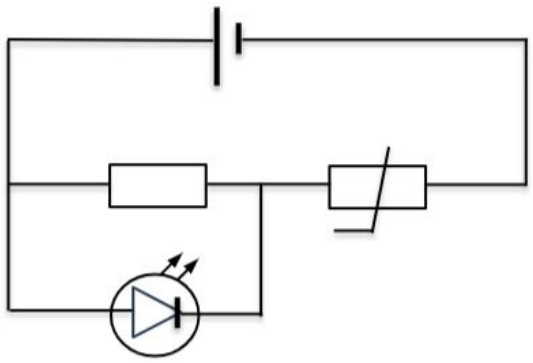
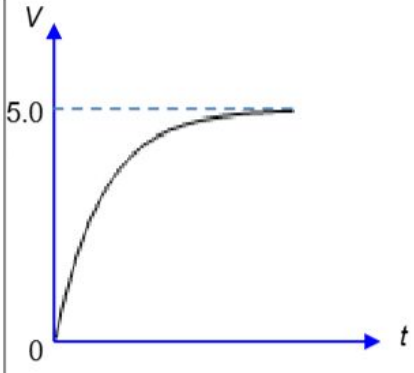
7	(a)	(i)	For the uniform wire, resistance of a segment of wire is proportional to the length of the segment. Hence by potential divider principle, $V_{ZY} = \frac{R_{ZY}}{R_{XY}} V_{XY} = \frac{L_{ZY}}{L_{XY}} V_{XY} = \frac{(6.0-2.0)}{6.0}(12.0)$ [C1] $= 8.0$ V [C1] Or $V_{XZ} = \frac{R_{XZ}}{R_{XY}} V_{XY} = \frac{L_{XZ}}{L_{XY}} V_{XY} = \frac{(2.0)}{6.0}(12.0)$ [C1] $= 4.0$ V [C1] Since point Y is earthed and hence at zero potential, potential at point Z = – 8.0 V [A1]
		(ii)	$R = \frac{\rho L}{A}$ where A is the cross-sectional area of the wire [M1] $A = \frac{\text{volume of wire}}{L}$ where volume of wire $= \frac{m}{D}$ [M1] Therefore $R = \frac{\rho L}{\left(\frac{m}{DL}\right)} = \frac{\rho D L^2}{m}$ (shown)
		(iii)	By $R = \frac{\rho D L^2}{m}$, since the length *L* is doubled while keeping the same cross-sectional area and material, the mass *m* will double. [M1] Or By $R = \frac{\rho L}{\text{A}}$, *L* is doubled while resistivity ρ and area A remains unchanged, [M1] Hence the <u>new resistance increases by a factor of 2</u> and the student is wrong. [A1]
	(b)	(i)	Increasing the p.d. across the thermistor results in <u>an increase in current, which in turn causes the temperature to rise</u>. [B1] The temperature rise causes <u>an increase in the number of (mobile) charge carriers</u>. [B1] The <u>effect of increase in number density of charge carriers outweighs the effect of increase in collisions between the free electrons and the lattice ions</u>. [B1] Hence, <u>resistance decreases</u> (i.e. the ratio of $\frac{V}{I}$ decreases). [B1]

		(ii)	 B1 for correct parallel circuit (diode direction is correct) B1 for correct circuit symbols (of LED, thermistor, cell and resistor) When temperature increases, the resistance of the thermistor decreases hence potential difference across the thermistor will decrease [M1] or When temperature increases, the resistance of the thermistor decreases hence the effective resistance of the circuit decreases, by potential divider principle, [M1] This causes the potential difference across the LED to increase beyond its threshold voltage hence turning it on. [A1]
		(iii)	The resistor is to prevent the LED from lighting up at low temperature. [B1]
	(c)	(i)	 M1 – correct shape A1 – correct label of 5.0 V
		(ii)	$V = V_0\left(1 - e^{-\frac{t}{RC}}\right)$ $2.0 = 5.0\left(1 - e^{-\frac{t}{3.0\times10^{6}(2.0\times10^{-6})}}\right)$ [M1] $t = 3.1\ \text{s}$ [A1]